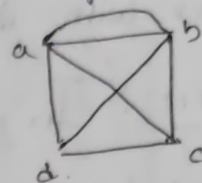
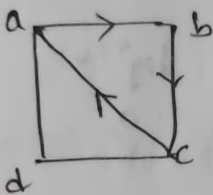


Graph Theory :-

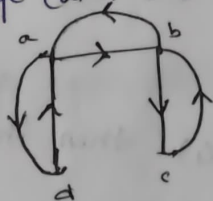
* **MultiGraph** :- A Graph containing more than one edge b/w the pair of vertices then it is called as Multi-Graph.



* **Mixed Graph** :- A Graph containing both directed and undirected edges.



* **Symmetric Directed Graph** :- A Graph $G(V, E)$ such that for every directed edge $(a, b) \in E$, there exist a directed edge $(b, a) \in E$.

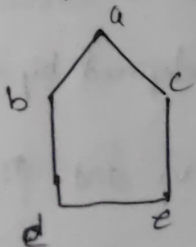


* **Simple Graph** :- a graph $G(V, E)$ with no self loops and parallel edges is called simple graph.



* **Order of the Graph** :- A Graph $G(V, E)$ with V as set of vertices then $|V(G)|$ or $|V|$ denotes the no. of vertices is called the order of the graph.

* **Size of the Graph** :- The no. of edges in the Graph G is called size of the Graph, denoted by $|E(G)|$ or $|E|$.



$$|V| = 5$$

$$|E| = 5$$

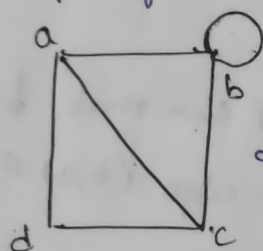
37/37

* Degree of the Graph :- Degree of vertex in an undirected graph is the no. of edges incident with it. A loop at the vertex contributes twice to the degree of that vertex.

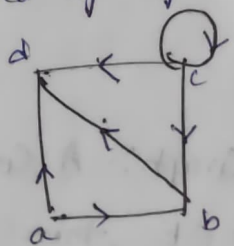
A vertex with degree zero is called an isolated vertex.

Indegree :- In a di-graph the no. of vertices incident to a vertex is called indegree of the vertex and it is denoted by $\deg^-(v)$.

Outdegree :- The no. of vertices incident from a vertex is called Outdegree of a vertex. It is denoted by $\deg^+(v)$.



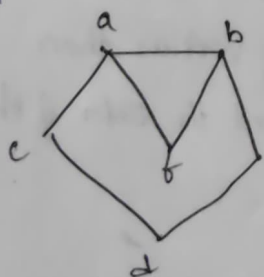
$a = 3$
 $b = 4$
 $c = 3$
 $d = 2$



$\deg^+(v)$ $\deg^-(v)$
 a 0 2
 b
 c
 d

Degree Sequence :-

If $v_1, v_2, \dots, v_n$ are the vertices of Graph G , then the sequence $d_1, d_2, \dots, d_n$ where $[d_i \leq d_{i+1}]$ d_i is degree of v_i called the degree sequence of G such that $d_1 \leq d_2 \leq d_3 \leq \dots \leq d_n$.



$d(a) = 3$ $d(e) = 2$
 $d(b) = 3$ $d(f) = 2$
 $d(c) = 2$
 $d(d) = 2$

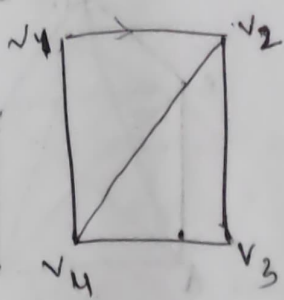
degree sequence = $\{2, 2, 2, 2, 3, 3\}$

Representation of Graph.

① Adjacency Matrix :- Let $G(V, E)$ be a graph with n vertices ordered from $v_1, v_2, \dots, v_n$. then the Matrix $A_G (n \times n)$ is defined by

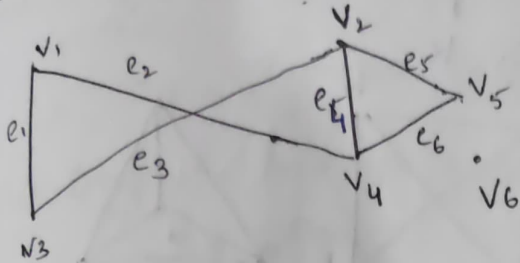
$A_G = [a_{ij}]_{n \times n}$ where $a_{ij} = 1$ if v_i is adjacent to v_j and $a_{ij} = 0$ otherwise.

-wise.



	v_1	v_2	v_3	v_4
v_1	0	1	0	1
v_2	1	0	1	1
v_3	0	1	0	1
v_4	1	1	1	0

Incidence Matrix



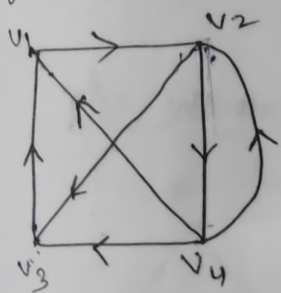
	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	0	1	1	1	0
v_3	1	0	1	0	0	0
v_4	0	1	0	1	0	1
v_5	0	0	0	0	1	1
v_6	0	0	0	0	0	0

⇒ Represented by $I_G = [a_{ij}]_{n \times m}$.

$a_{ij} = 1$ if v_i is incident on v_j other wise

③ Path Matrix :- A path matrix with all one's is called a fully connected graph.

If a column has all zero's then the indegree of that vertex is 0.



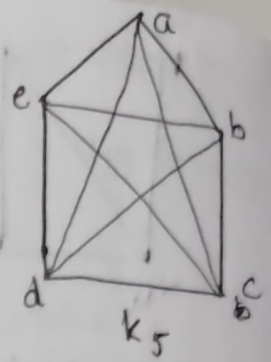
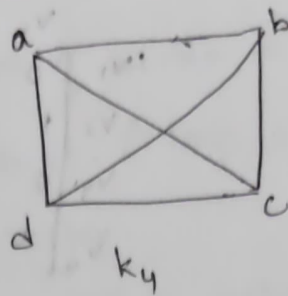
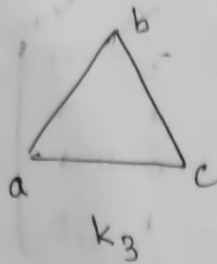
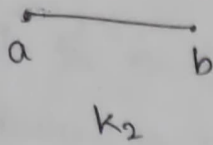
	v_1	v_2	v_3	v_4
v_1	1	1	1	1
v_2	1	1	1	1
v_3	1	1	1	1
v_4	1	1	1	1

Types of Graphs :-

* Complete Graph :- A graph with n vertices is said to be complete graph if it contains n mutually adjacent vertices.

→ K_5 graph is also called Kuratowski's first graph.

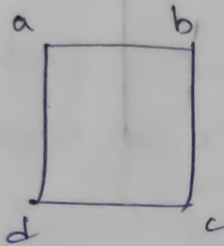
→ The degree of a complete graph is $n-1$.



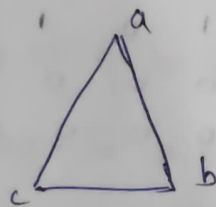
②

* k-regular Graph:- A graph G is k -regular graph if all its Vertices degree is k .

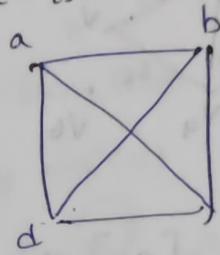
Eg:-



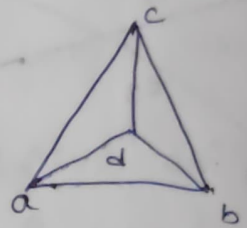
2-regular graph



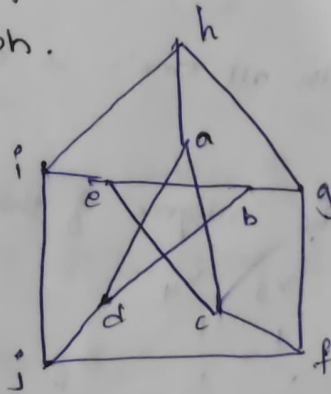
2 regular graph



3-regular graphs

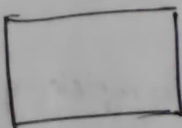


→ A 3-regular graph with 10 vertices and 15 edges is called Peterson Graph.



③

* Cycle Graph:- A Graph G is said to be Cyclic if all the edges form one cycle of length n and is denoted by C_n



C_4

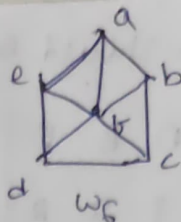
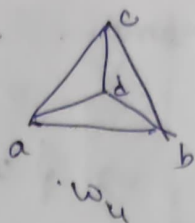
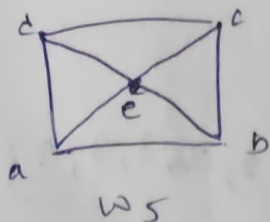


C_3



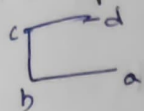
C_5

* ④ Wheel Graph:- A Wheel graph is obtained by adding a single new vertex to the each vertex of a circular graph. It denoted by W_n .

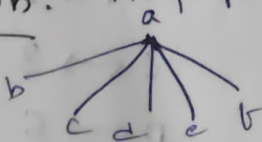


Null Graph :- A graph of order n with n vertices and no edges is called Null graph.

Path Graph :- Path graph is obtained by removing only one edge from Complete graph.



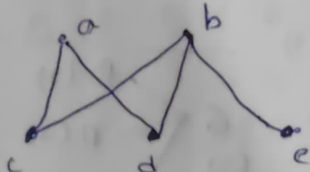
Star Graph :- A graph of the form $K_{1,n}$ is called star graph.



Bipartite Graph :- A graph which the vertex set can be partitioned into two sets m and n . In such a way the edge in G joins a vertex in m to a vertex in n .

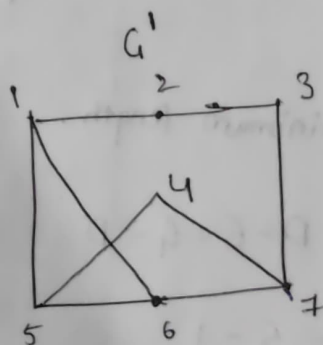
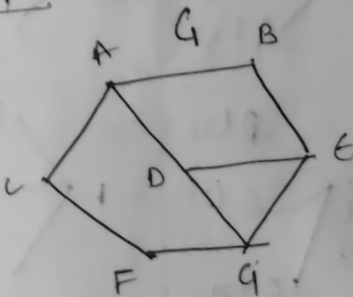
$m(a, b)$

$n(c, d, e)$



$$m \cap n = \emptyset$$

Graph Isomorphism :-



For two graphs to be isomorphic

$$G \cong G'$$

(i) No. of vertices should be equal. $|V(G)| = 7, |V(G')| = 7 \therefore |V(G)| = |V(G')|$

(ii) No. of edges should be equal $|E(G)| = 9, |E(G')| = 9, \therefore |E(G)| = |E(G')|$

(iii) Degree Sequence should be same.

$G \Rightarrow$

	A	B	C	D	E	F	G
Degree	3	2	2	3	3	2	3

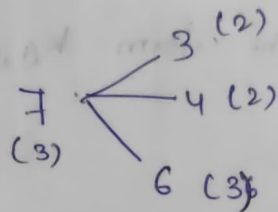
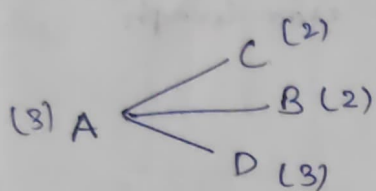
$\Rightarrow \{2, 2, 2, 3, 3, 3, 3\}$

$G' \Rightarrow$

	1	2	3	4	5	6	7
Degree	3	2	2	2	3	3	3

$\Rightarrow \{2, 2, 2, 3, 3, 3, 3\}$

(iv) One-to-one mapping.

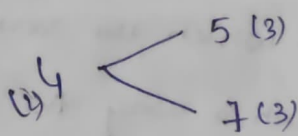
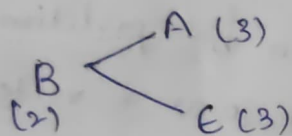


so, $A \leftrightarrow 7$

$B \leftrightarrow 4$

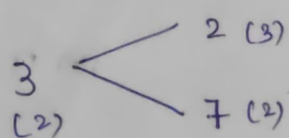
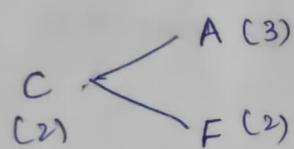
$C \leftrightarrow 3$

$D \leftrightarrow 6$

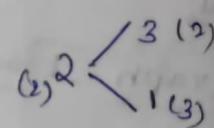
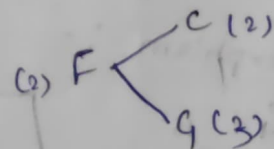
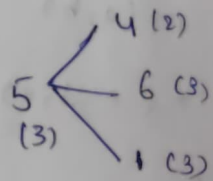
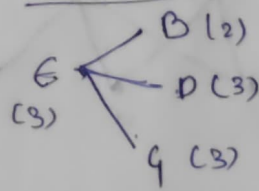
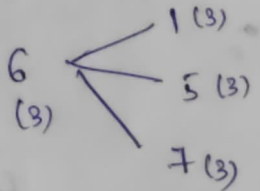
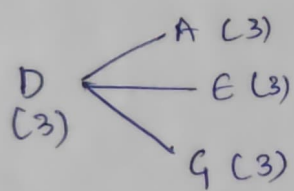


$E \leftrightarrow 5$

$F \leftrightarrow 2$



$G \leftrightarrow 1$



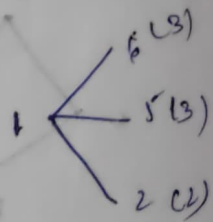
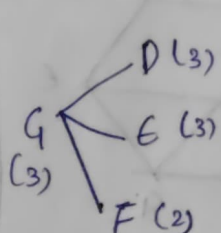
(v) Cycle of minimum length.

$G \Rightarrow$ ~~A-B-A~~ D-E-G-D

len = 3

$G' \Rightarrow$ 1-6-5-1

len = 3



(vi) edge preservation.

A - B

B - E

E - G

G - F

F - C

C - A

A - D

D - E

E - G

7 - 4

4 - 5

5 - 1

1 - 2

2 - 3

3 - 7

7 - 6

6 - 5

5 - 1

(vii) Adjacency Matrix.

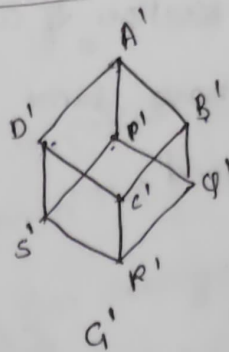
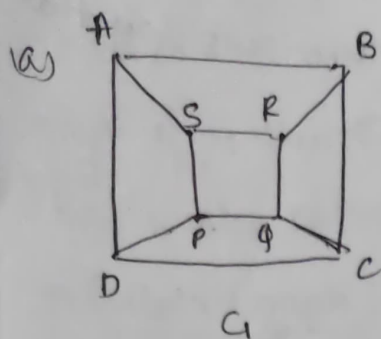
$$G = \begin{matrix} & \begin{matrix} A & B & C & D & E & F & G \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \\ G \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$G' = \begin{matrix} & \begin{matrix} 7 & 4 & 3 & 6 & 5 & 2 & 1 \end{matrix} \\ \begin{matrix} 7 \\ 4 \\ 3 \\ 6 \\ 5 \\ 2 \\ 1 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$\therefore G = G'$$

$$\therefore G \cong G' \text{ (G is isomorphic to G')}$$

Q Prove Graph Isomorphism:-



(1) $V(G) = 8$

$V(G') = 8$

$V(G) = V(G')$

(2) $E(G) = 12$

$E(G') = 12$

$E(G) = E(G')$

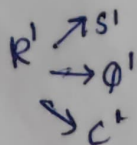
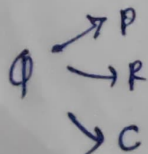
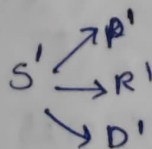
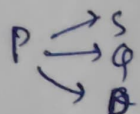
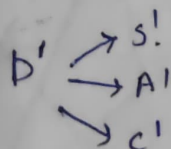
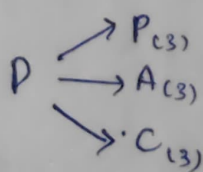
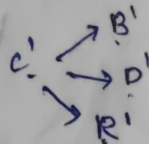
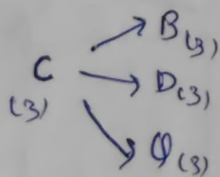
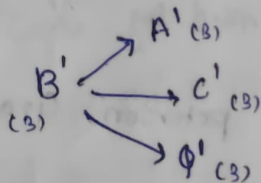
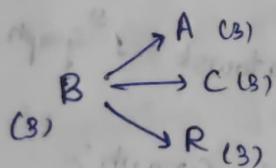
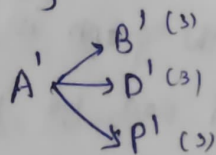
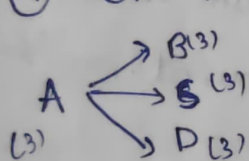
(3) Degree Sequence

G : A B C D P Q R S
3 3 3 3 3 3 3 3

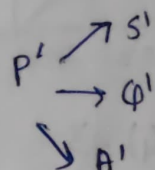
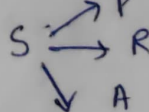
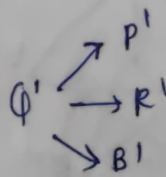
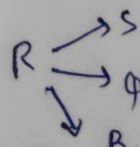
G' : A' B' C' D' P' Q' R' S'
3 3 3 3 3 3 3 3

$= \{3, 3, 3, 3, 3, 3, 3, 3\} = \{3, 3, 3, 3, 3, 3, 3, 3\}$

(4) One to one mapping



$A \leftrightarrow A'$	$A \mapsto A'$
$B \leftrightarrow B'$	$B \mapsto B'$
$D \leftrightarrow P'$	$C \leftrightarrow C'$
$S \leftrightarrow P'$	$D \leftrightarrow D'$
$C \leftrightarrow C'$	$P \leftrightarrow S'$
$R \leftrightarrow Q'$	$Q \leftrightarrow R'$
$Q \leftrightarrow R'$	$R \leftrightarrow Q'$
$P \leftrightarrow S'$	$S \leftrightarrow P'$



(i) cycle of min length :-

$G \rightarrow A-B-C-D-A$
 $G' \rightarrow A'-B'-C'-D'-A'$

(ii) edge preservation :-

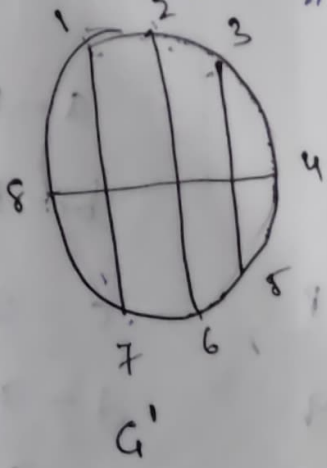
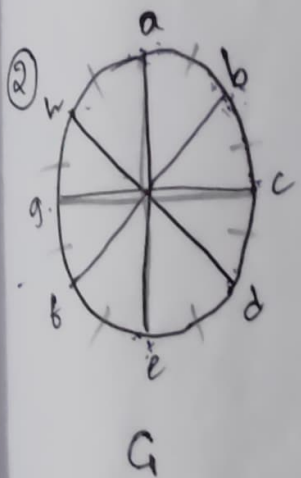
$A-B$ $A'-B'$
 $B-C$ $B'-C'$
 $C-D$ $C'-D'$
 $D-P$ $D'-S'$
 $P-Q$ $S'-R'$
 $Q-R$ $R'-\phi'$
 $R-S$ $\phi'-P'$ $P'-A'$
 $S-A$

(iii) Adjacency Matrix :-

	A	B	C	D	P	Q	R	S
A	0	1	0	1	0	0	0	1
B	1	0	1	0	0	0	1	0
C	0	1	0	1	0	1	0	0
D	1	0	1	0	1	0	0	0
P	0	0	0	1	0	0	1	1
Q	0	0	1	0	0	0	1	0
R	0	1	0	0	0	1	0	1
S	1	0	0	0	1	0	1	0

	A'	B'	C'	D'	S'	R'	Q'	R'
A'	0	1	0	1	0	0	1	0
B'	1	0	1	0	0	1	0	0
C'	0	1	0	1	0	0	0	0
D'	1	0	1	0	1	0	0	0
S'	0	0	0	1	0	0	1	0
R'	0	0	1	0	1	1	0	1
Q'	0	1	0	0	0	0	1	0
P'	1	0	0	0	0	0	1	0

$\therefore G \cong G'$



① $V(G) = 8$, $V(G') = 8$ $\therefore V(G) = V(G')$

② $E(G) = 12$, $E(G') = 12$ $\therefore E(G) = E(G')$

③ $G \rightarrow$ Degree sequence : $\{3, 3, 3, 3, 3, 3, 3, 3\}$
 $a \ b \ c \ d \ e \ f \ g \ h$
 $3 \ 3 \ 3 \ 3 \ 3 \ 3 \ 3 \ 3$

$G' \rightarrow$ Degree sequence : $\{3, 3, 3, 3, 3, 3, 3, 3\}$
 $1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8$
 $3 \ 3 \ 3 \ 3 \ 3 \ 3 \ 3 \ 3$

$1 \leftrightarrow a$

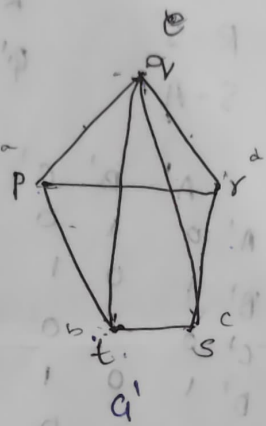
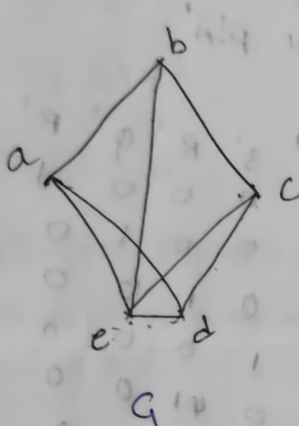
② Cycle of min length:-

In graph G cycle is:- $a-b-c-d-e-a = 5$

G' cycle is:- $3-4-5-3 = 2$

$\therefore G \neq G'$ as not Isomorphic to G'

①



① No. of Vertices

$$V(G) = 5$$

$$V(G') = 5$$

② No. of Edges

$$E(G) = 8$$

$$E(G') = 8$$

$$E(G') = 8$$

③ Degree Sequence

G :- $a \ b \ c \ d \ e$

$3 \ 3 \ 3 \ 3 \ 4$

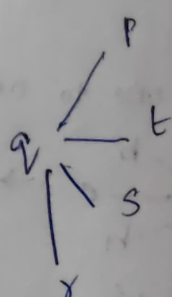
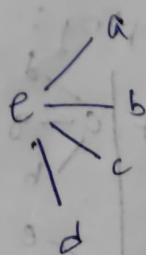
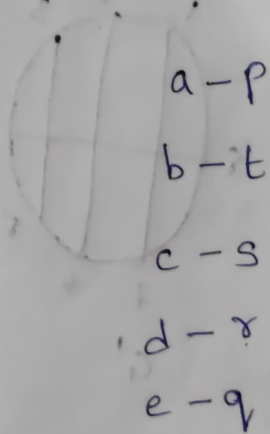
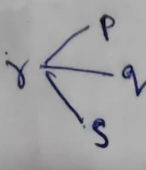
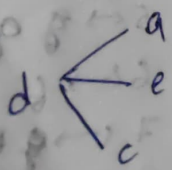
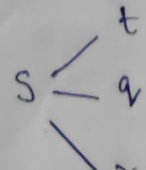
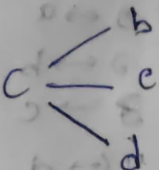
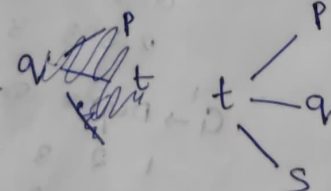
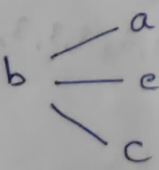
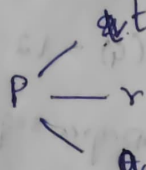
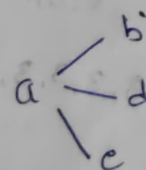
$= \{3, 3, 3, 3, 4\}$

G' :- $p \ q \ r \ s \ t$

$3 \ 4 \ 3 \ 3 \ 3$

$= \{3, 3, 3, 3, 4\}$

④ One to One mapping:-



(v) Cycle of min length:-

$G:- a \rightarrow r \rightarrow s \rightarrow q \rightarrow e$
 $G':- e \rightarrow d \rightarrow c \rightarrow e$

edge preservation:-

$a-b$ $p-t$

$b-c$ $t-s$

$c-d$ $s-r$

$d-e$ $r-q$

$e-a$ $q-p$

$a-d$ $p-r$

$e-c$ $q-s$

$e-b$ $q-t$

(vii) Adjacency Matrix:-

	a	b	c	d	e
a	0	1	0	1	1
b	1	0	1	0	1
c	0	1	0	1	1
d	1	0	1	0	1
e	1	1	1	1	0

G'

	p	t	s	r	q
p	0	1	0	1	1
t	1	0	1	0	1
s	0	1	0	1	1
r	1	0	1	0	1
q	1	1	1	1	0

$$G \cong G'$$

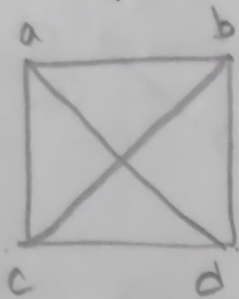
//

$$[0 \ 0 \ 0]$$

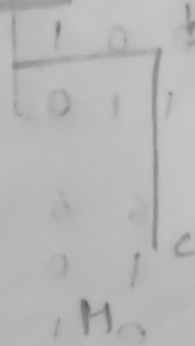
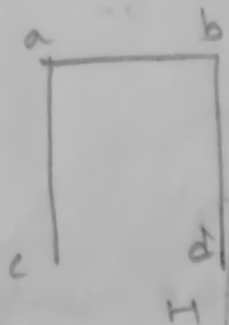
$$[0 \ 0 \ 0]$$

Subgraph: - let G be a graph, we say H as a subgraph of G if and only if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$

Graph G



Subgraphs

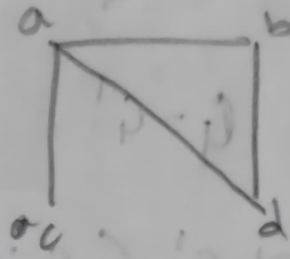
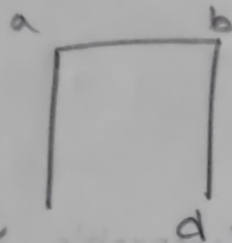


Spanning sub-graph: - a Graph H is spanning sub-graph of G

iff (i) $V(H) = V(G)$

$E(H) \subseteq E(G)$

Spanning subgraphs

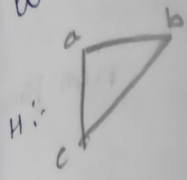


Induced Subgraph: Let G be a graph and W be a vertex set which is sub-set of G then Sub-graph induced by vertex set W is given by

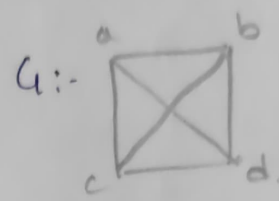
$$V(H) = W$$

$E(H)$ = All edges that are adjacent within vertices of W .

$$W = \{a, b, c\}$$



$$V(H) = W$$

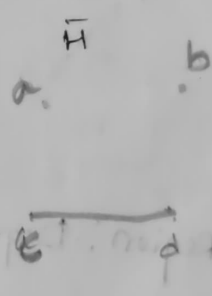
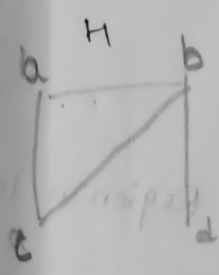
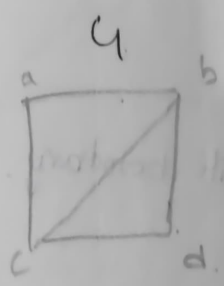


Complement of Sub-graph:

The complement of sub-graph H in graph G is given by $\bar{H}$ where

$$V(\bar{H}) = V(H)$$

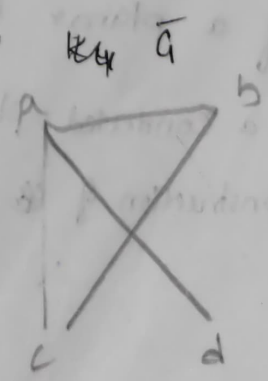
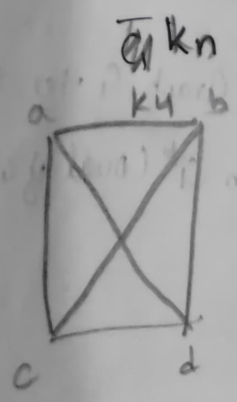
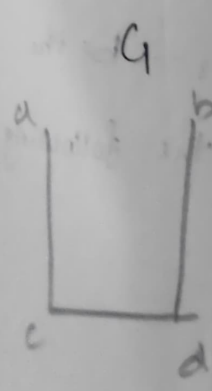
$$E(\bar{H}) = \{(x,y) / (x,y) \notin H \text{ and } (x,y) \in G\} = E(G) - E(H)$$



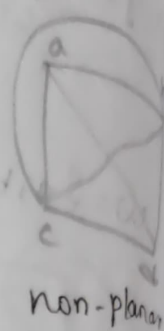
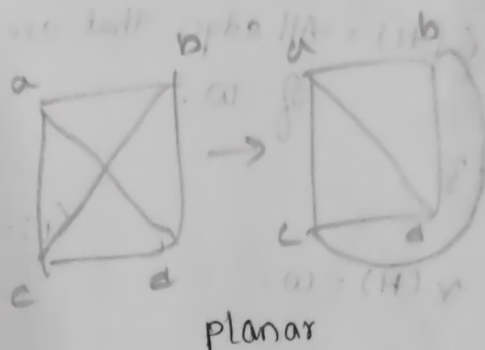
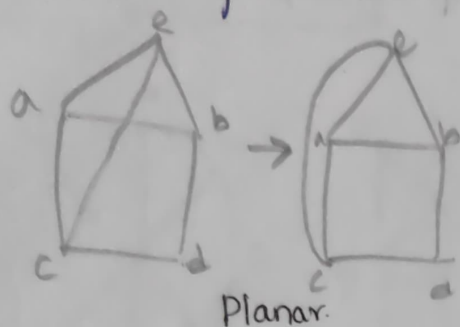
Complement of Graph: Complement of graph G of order n is equivalent to the complement of G in K_n graph.

The complement of Graph G , $\bar{G}$ is given by $V(\bar{G}) = V(G)$

$$E(\bar{G}) = \{(x,y) / (x,y) \notin G\}$$

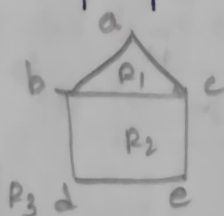


Planar Graphs :- A Graph G is called a planar graph if the edges are not crossing with each other, i.e. the edges meet only at vertices:



Region of a Graph :-

A region of a planar graph is defined to be an area of the plane i.e. bounded by edges and further can not be divided.

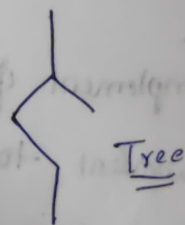


Degree of Region :- Degree of Region is length of its boundary.

Degree of $R_1 = 3$

$R_2 = 4$

$R_3 = 5$



* A graph with region 1 is called a Tree.

Dual of a planar Graph.

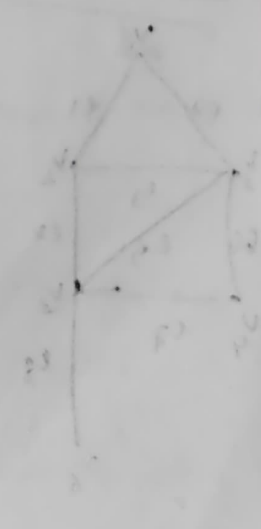
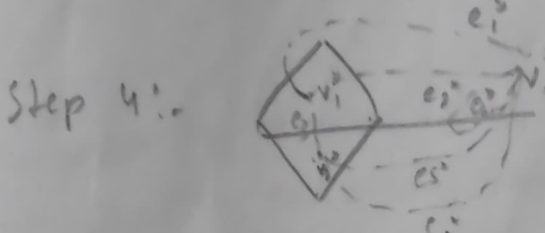
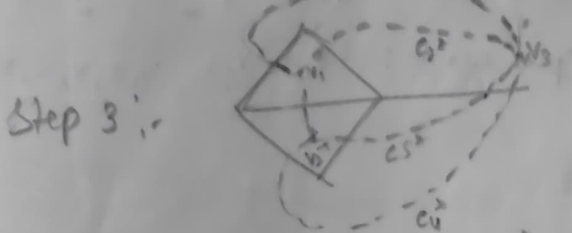
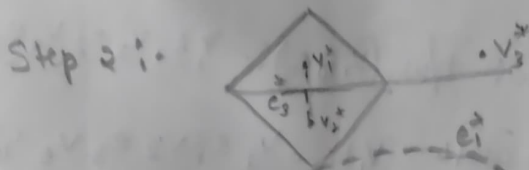
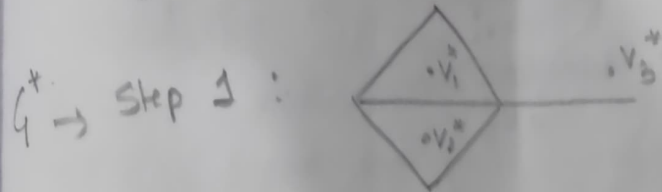
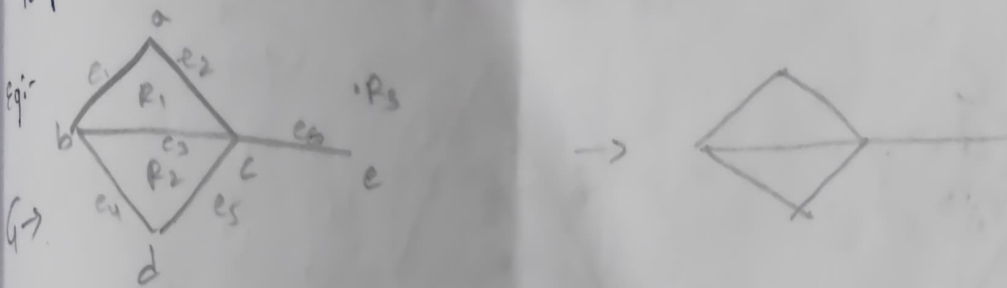
Consider a connected planar Graph G . let $R_1, R_2, R_3, \dots$ be the regions. Construction of Graph G^* (Dual of G), has the following steps.

① Choose one point inside each of regions R_1, R_2, R_3 and so on and denote these points by $v_1^*, v_2^*, v_3^* \dots$ these are the new vertices of G^* .

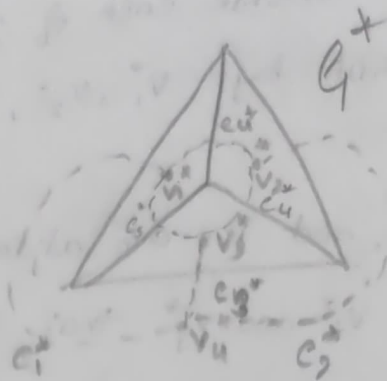
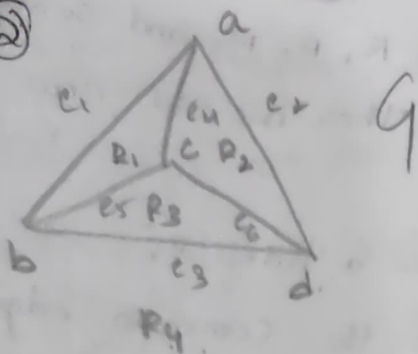
② If two regions R_i and R_j are adjacent draw a line e_k^* joining the points v_i^* and v_j^* that intersects the common edge e_k exactly once.

③ If there is more than one edge common to R_i and R_j draw one line e_p^* b/w the points for each common edge e_p intersecting e_p exactly once.

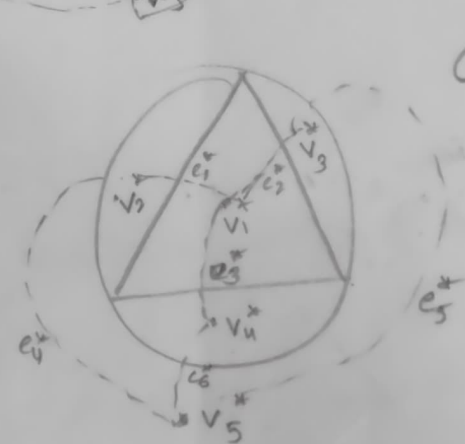
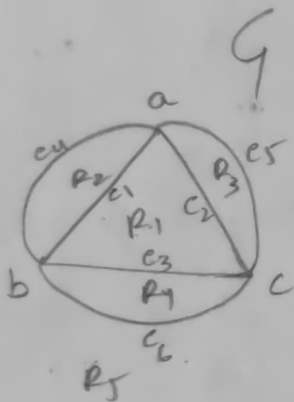
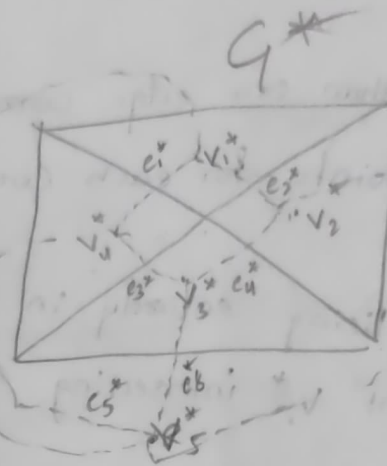
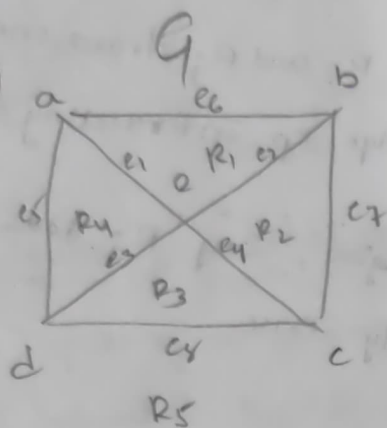
④ For each edge e_i lying entirely in one region, say R_i draw a loop e_i^* at the point v_i^* intersecting e_i exactly once.



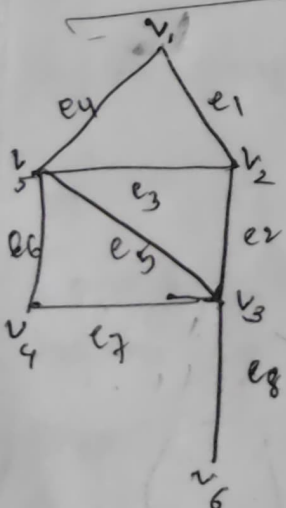
②



③



Euler and Hamilton graph:-



Graph G
for walk

walk :- b/w (v_1, v_3) :- $v_1, e_1, v_2, e_2, v_3, e_8, v_6$ $l=3$

closed walk $:- (v_1) :- v_1, e_1, v_2, e_3, v_5, e_4, v_1$ $l=3$

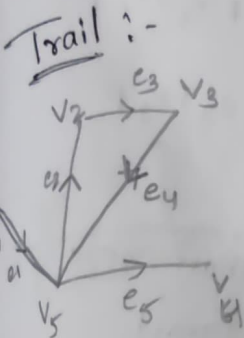
Open walk :- $(v_1, v_3) :- v_1, e_1, v_2, e_2, v_3, e_8, v_6$ $l=3$

Walk :- Consider graph G having atleast one edge, a sequence of vertices which begins and ends with vertices is called a walk.

Closed walk :- A walk that begins and ends at the same vertex is called closed walk.

Open walk :- A walk that begins and ends at two different vertices is called an Open Walk.

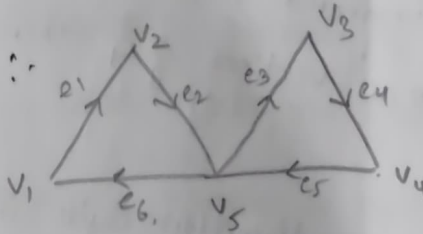
Trail :-



v_1 to v_4 :- $v_1 \underline{e_1} v_2 \underline{e_2} v_3 \underline{e_3} v_4 \underline{e_4} v_5 \underline{e_5} v_4$

An open walk with no edges appear more than once then the walk is called a trail.

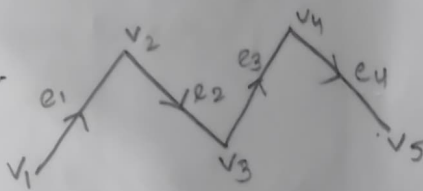
Circuit :-



v_1 to v_1 :- $v_1 \underline{e_1} v_2 \underline{e_2} v_3 \underline{e_3} v_4 \underline{e_4} v_5 \underline{e_5} v_1 \underline{e_6} v_1$

A closed walk in which no edge appears more than once. Then it is called circuit.

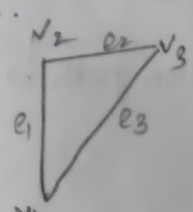
Path :-



v_1 to v_5 :- $v_1 \underline{e_1} v_2 \underline{e_2} v_3 \underline{e_3} v_4 \underline{e_4} v_5$

A trail in which no vertex appears more than once is known as path.

Cycle :-



v_1 to v_1 :- $v_1 \underline{e_1} v_2 \underline{e_2} v_3 \underline{e_3} v_1$ (Ignore v_1)

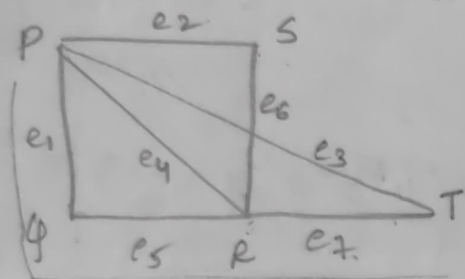
A circuit in which the terminal vertex does not appear as an internal vertex and no two internal vertices are repeated is called a Cycle.

Euler's Graph :-

→ Euler's circuit :- If there is a circuit in G which contains all edges of G then that circuit is called Euler circuit in G .

This is also called as Eulerian line / Euler tour.

→ Euler's Graph :- A graph that contains an Euler circuit then it is called as Euler's Graph.



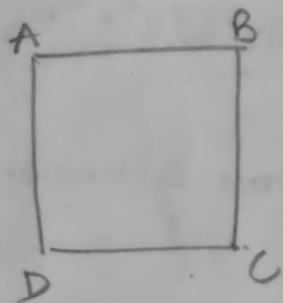
Circuit :- $P \xrightarrow{e_1} Q \xrightarrow{e_7} R \xrightarrow{e_2} S \xrightarrow{e_6} T \xrightarrow{e_3} P$
 $P \xrightarrow{e_1} Q \xrightarrow{e_5} R \xrightarrow{e_4} P \xrightarrow{e_2} S \xrightarrow{e_6} R \xrightarrow{e_7} T \xrightarrow{e_3} P$

Trail :-

Euler's Trail :- If there is a trail in graph G that contains all the edges of ' G '. Then the trail is called Euler's Trail.

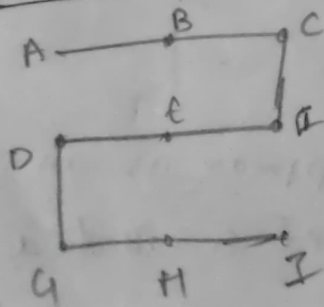
Hamilton Cycle :- Let G be a connected graph ; if there is a cycle in graph ' G ' that contains all vertices of G then it is called as Hamilton Cycle.

Hamilton Graph :- A graph that contains Hamilton Cycle is called Hamilton Graph.



cycle :- $A-B-C-D-A$.

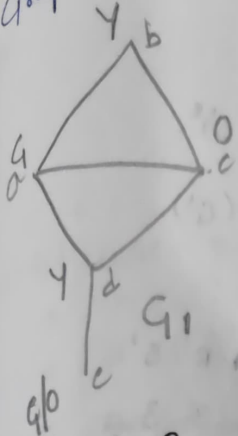
Hamilton Graph path :- A path in a graph which includes every vertex of a graph is called Hamilton path in a graph.



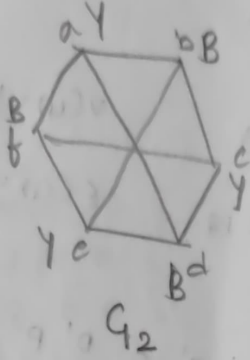
path :- $A-B-C-D-E-F-G-H-I$

Graph Colouring :-

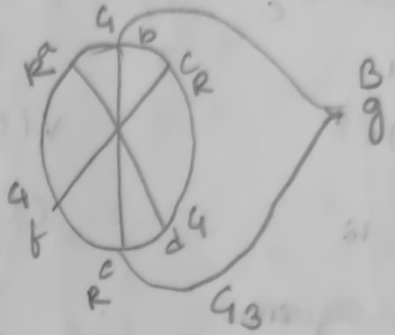
If we assign colours to vertices of a Graph in such a way that no two vertices should have same colour. Then we say that Graph G is properly coloured.



$$\chi(G_1) = 3$$



$$\chi(G_2) = 2$$



$$\chi(G_3) = 3$$

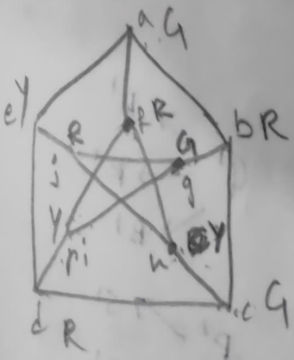
Chromatic Number :- A Graph G is said to be K -colourable if we can properly colour it with K colors.

The ~~chromatic~~ Chromatic Number of the graph is the minimum number of colours with which the graph can be properly coloured.

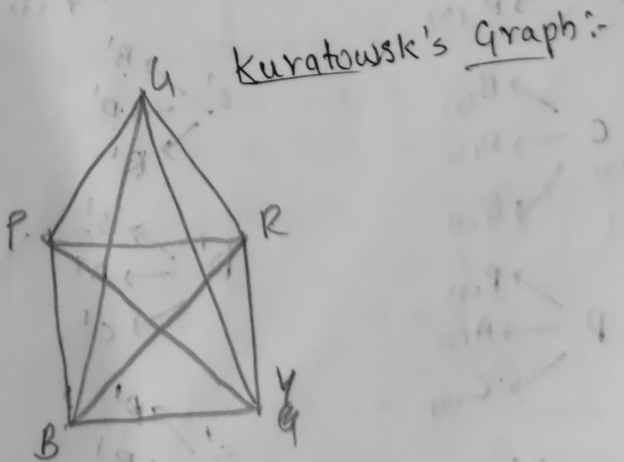
The Chromatic Number is represented by $\chi(G)$.

Q) Find the Chromatic Number of Peterson Graph Kuratowski's 1st graph.

Q) Peterson Graph :-



$$\chi(G) = 3$$



$$\chi(G) = 5$$